STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	I-III Semester (2025-26)
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Branch	CSE
Course Outcome	4.

Work Done Summary

In this case study, we analyzed Chennai Metro fare optimization using Dijkstra's Algorithm with multi-criteria routing.

The work focused on:

1. Representing metro stations as graph vertices.
2. Representing metro connections as weighted edges.
3. Using fare as the primary optimization parameter.
4. Using travel time as a tiebreaker.
5. Applying Dijkstra's shortest path algorithm.
6. Using lexicographic comparison `(fare, time)`.
7. Finding the cheapest path from ALD to WMN.
8. Comparing multiple candidate routes.
9. Calculating cumulative fare and travel time.
10. Demonstrating Pareto-optimal routing concepts.

Implementation Details

1. Modeled metro stations as graph nodes.
2. Stored edges with two weights:
    - Fare (₹)
    - Travel time (minutes)
3. Used adjacency lists for graph representation.

4. Initialized distance array as:

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(fare, time)
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5. Used a PriorityQueue (min-heap) ordered by:
    - Minimum fare first
    - Minimum time second
6. Started Dijkstra traversal from ALD station.
7. Extracted the node with smallest `(fare, time)` pair.
8. Relaxed neighboring edges using lexicographic comparison.
9. Updated shortest paths only when:
    - Fare becomes smaller OR
    - Fare same but time smaller
10. Maintained cumulative fare and travel time during traversal.
11. Identified the cheapest and fastest valid route to WMN.
12. Demonstrated multi-criteria shortest path optimization.

RESULTS:SCREENSHOTS/Output:

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Starting Dijkstra from ALD...

Relaxed Paths:

ALD → AVN : Fare = 10 , Time = 4
ALD → GTM : Fare = 8 , Time = 7

GTM → MGR : Fare = 18 , Time = 12

AVN → CMR : Fare = 22 , Time = 10

MGR → WMN : Fare = 30 , Time = 16

CMR → TNB : Fare = 37 , Time = 16

Final Shortest Path:

ALD → GTM → MGR → WMN

Minimum Fare = ₹30
Total Travel Time = 16 minutes

Traversal Completed Successfully.
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Conclusion

In this case study, Dijkstra’s Algorithm was used because it efficiently finds the shortest path in weighted graphs such as metro transportation networks.

Fare was treated as the primary optimization parameter, while travel time acted as a secondary tiebreaker using lexicographic comparison. This ensured that passengers always received the cheapest route, and among equal-fare routes, the fastest one was selected.

The PriorityQueue-based implementation improved efficiency by always processing the currently optimal route first. This approach is highly useful in real-world applications such as metro systems, navigation services, ride-sharing platforms, and intelligent transport systems where multiple optimization criteria must be considered simultaneously.

Github Link: https://github.com/ampssubrahmanyasreepad-svg/DSA-CaseStudies-correct

Student Signature	Faculty Signature
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